



# Basics of Electrical Quantities & Measurement Using

## KEYSIGHT INSTRUMENTS

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# **ELECTRICITY**



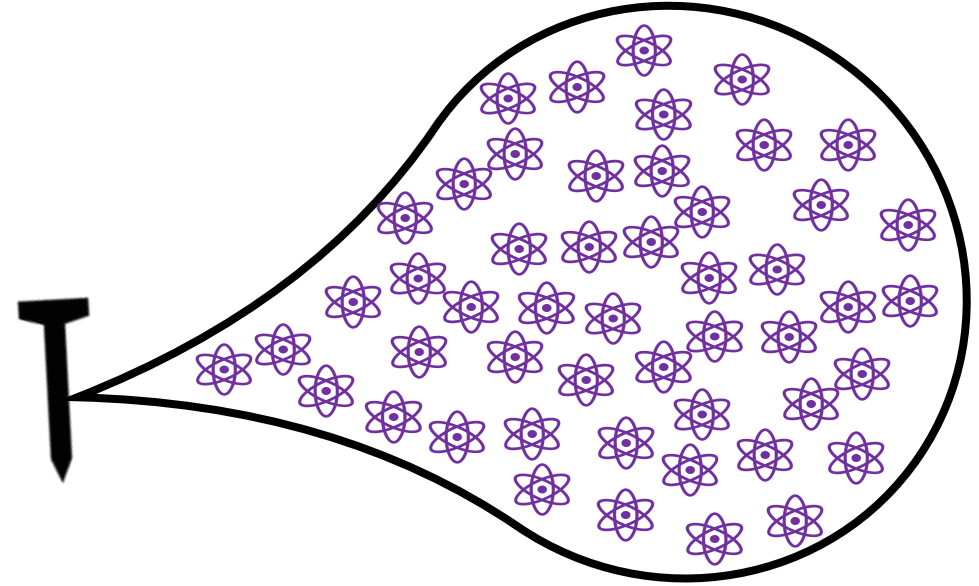
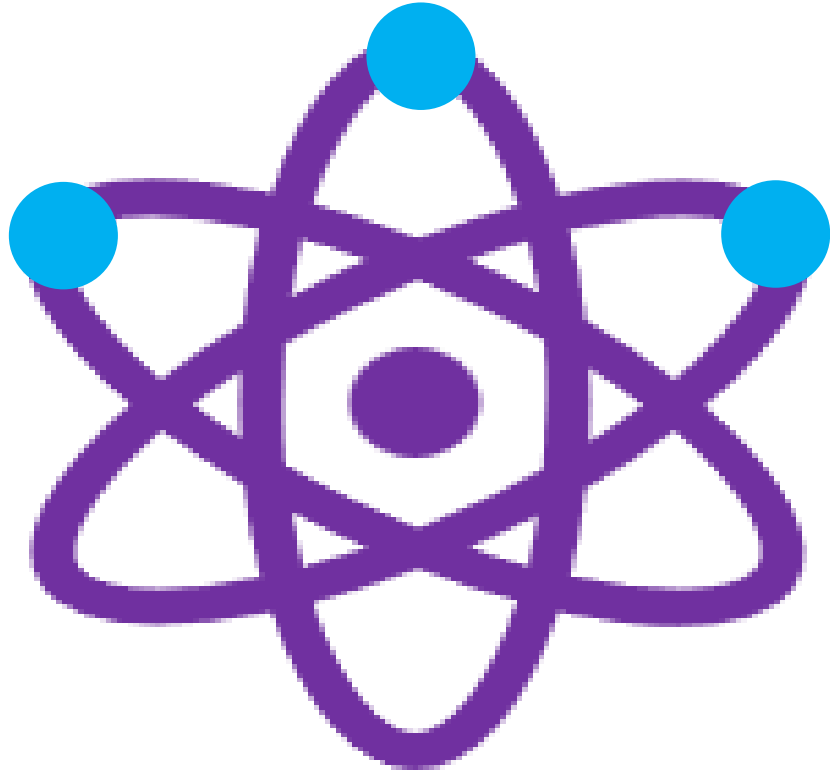
**Voltage**



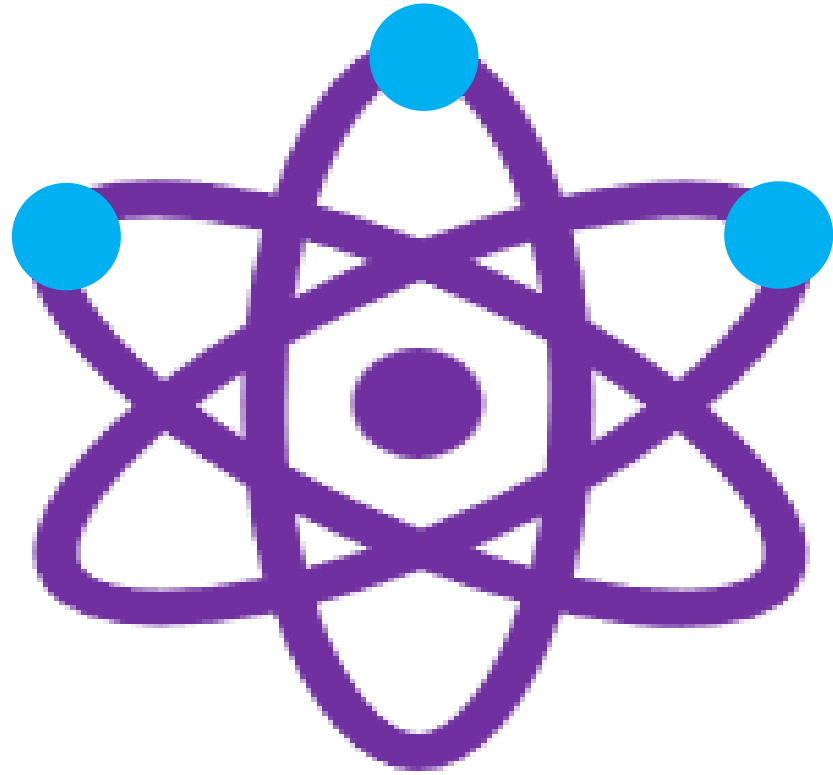
**Current**

**Do you truly know what  
they're all about ?**

**Everything around us  
is made of atoms**



# Everything around us is made of atoms



## ● **Electrons**

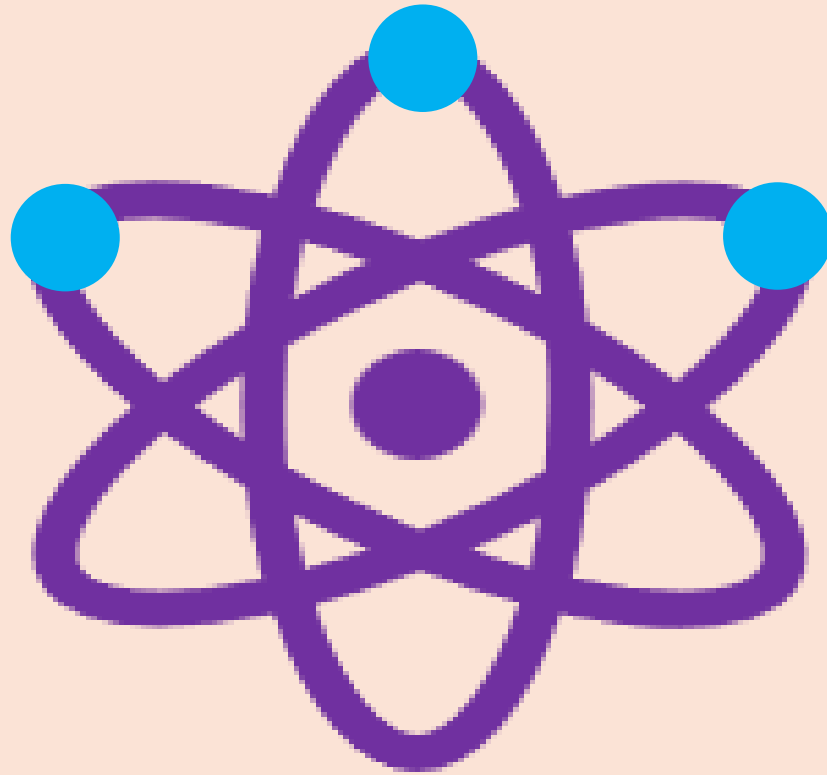
**Negatively Charged**

**Attracts Positive Charges**

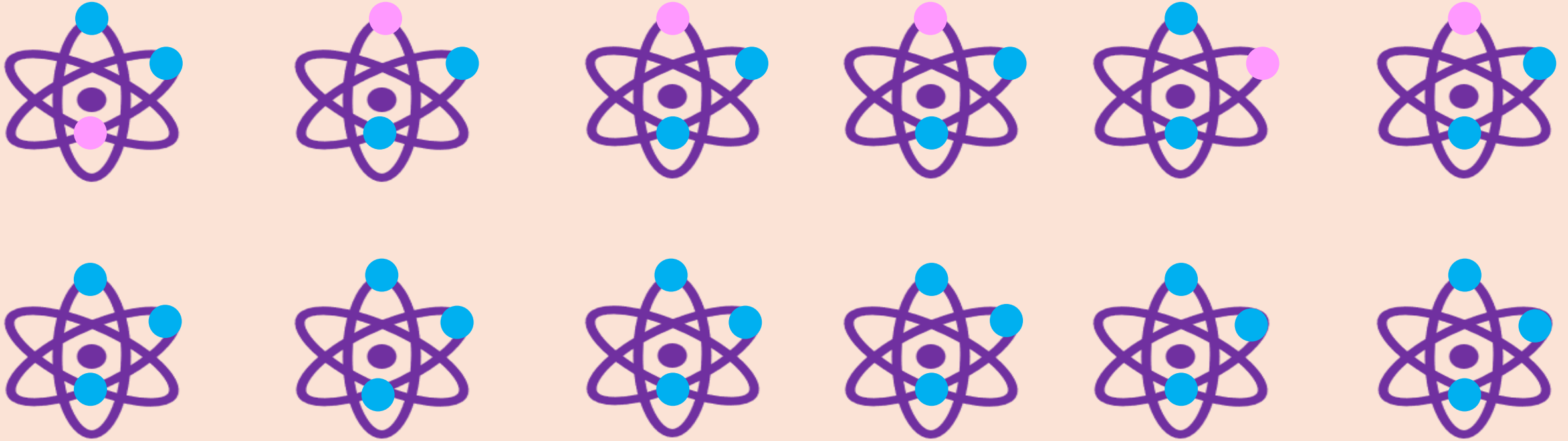
**Repels Negative charges**

**Orbits around nucleus**

**Some materials let these electrons break  
free from atoms and move through**



# Conductors



**There are trillions of free-flowing electrons in a small conducting materials**

# Conductors

**Materials that allow electrons to flow through them easily.**



**Ex : Iron, Copper, Gold, Etc..**

# Insulators

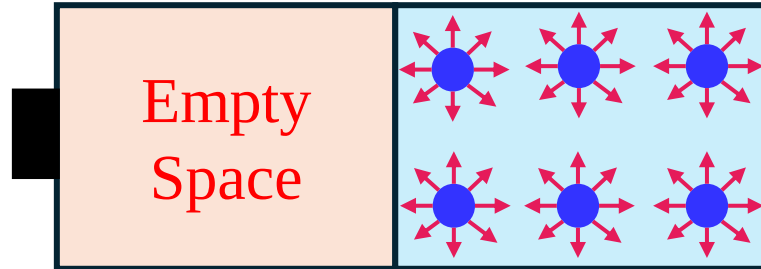
**Materials that do not allow electrons to flow through them.**



**Ex : Rubber, Plastic, glass, Etc..**

# Battery

**The Positive side has a huge empty space that electron can fill**

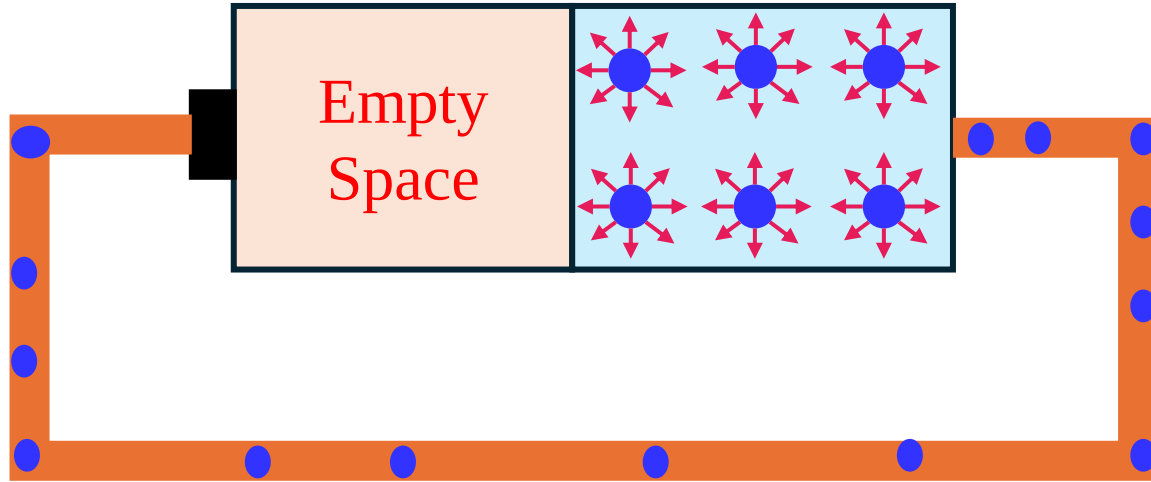


**The Negative side contains a lot of Electrons**

**Battery have two sides  
Positive (+) & Negative (-)**

- **The electrons in the negative side repel each other, it creates a huge electron pressure.**
- **The electrons are very much likely to move into the positive side that has less electron pressure**
- **If we consider battery alone there is no way to release this pressure to the positive side**

# Battery

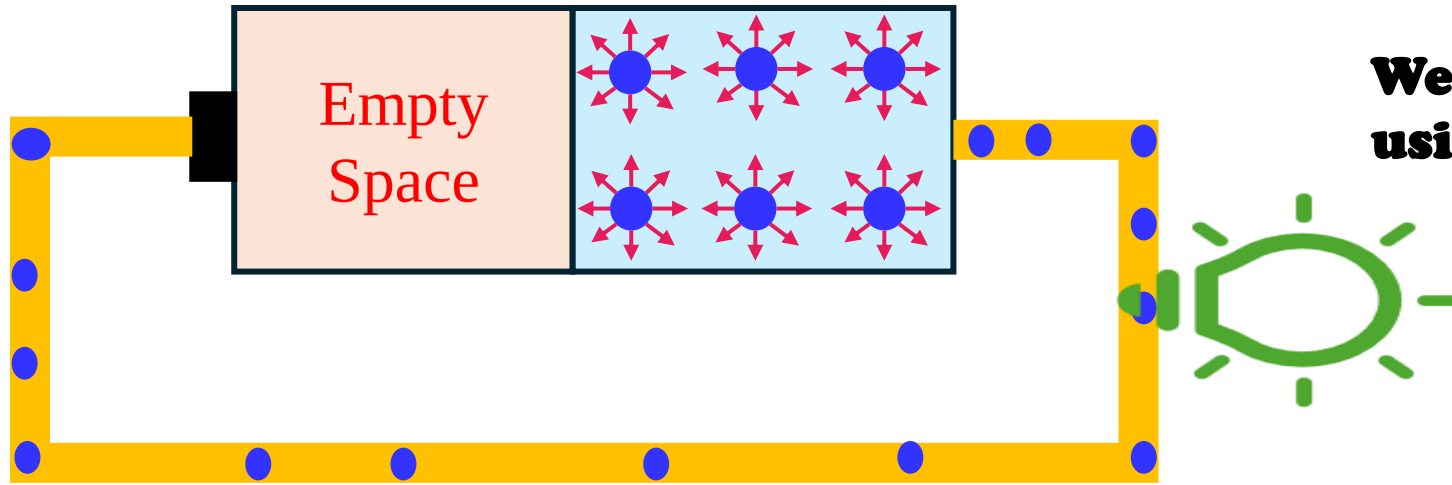


**Copper  
Wire**

**If we connect the copper wire  
like this in battery**

**The copper conductor makes a path for electrons to flow from the high-pressure  
negative side into the low-pressure positive side.**

**We are giving a path for electrons to flow because we need to get some work done  
their movement.**



**We are powering up our applications using the movement of electrons.**

**When Electrons move in one direction we call it electric current.**

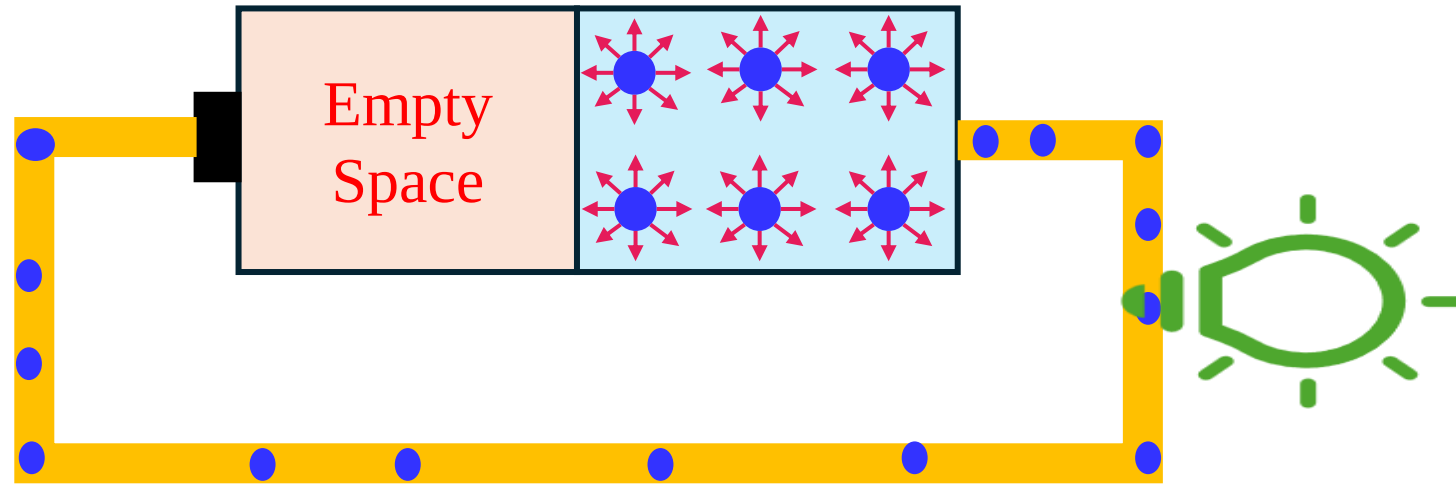
**Current**

**Electric current : A Flow of Electrons**

**But the direction of current is opposite to the direction of electron flow**

**Because early scientists assumed positive charges moved, and this convention was kept for consistency in theory and practice.**

# Current

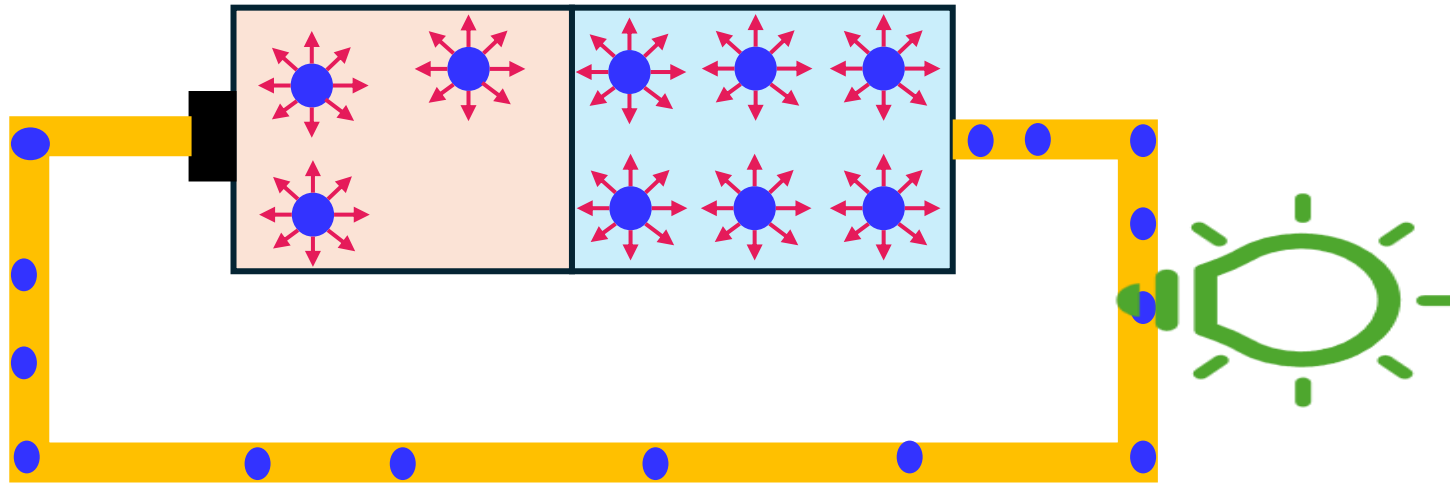


 **Electron Flow**

**Current** 

**We measure how many electrons pass  
by a spot in 1 second using a unit  
called **"Ampere" (A)**.**

**1 Ampere =  $6.24 \times 10^{18}$  electrons move  
in a second**



**At the beginning the electron pressure of the negative side is high  
& the positive side is Low**

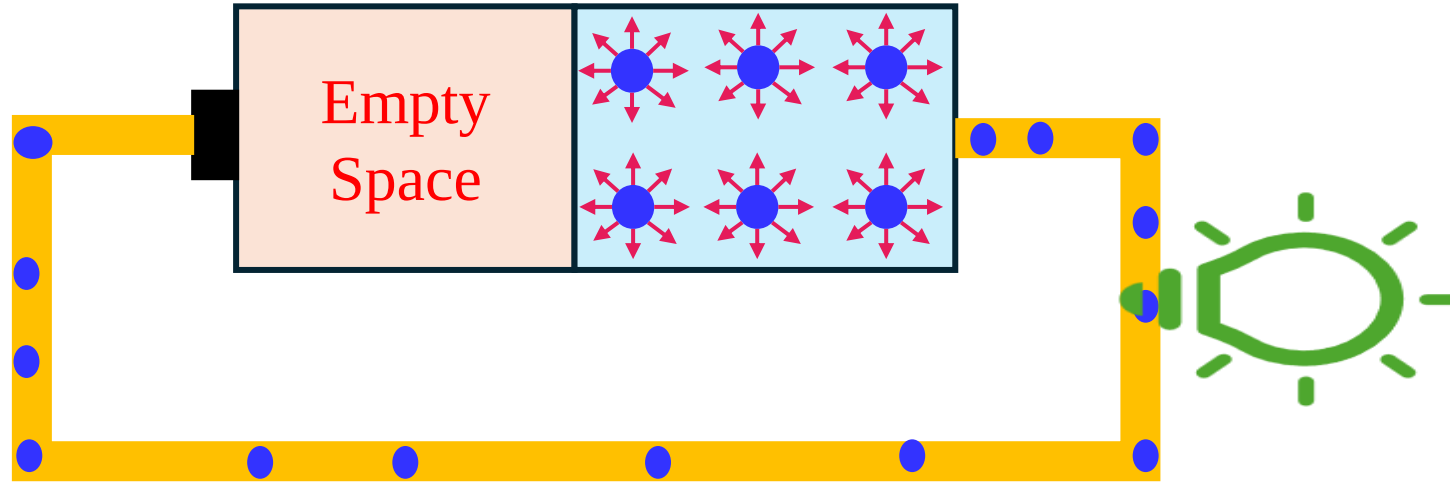
**▼ When the pressure of the negative side is gradually  
the electrons stop.**

**+** 

**▲ The Battery has drained out ...!**  
**The pressure of the Positive side is gradually  
increasing.**

**—** 

# Voltage



**Electrons Flow from high pressure to low pressure**

**To start an electric current, an electron pressure difference is needed.**

**Electron pressure = Electric potential.**

**The potential difference is measured by the unit called “voltage” (V)**

# **Current**



**Alternating  
Current**

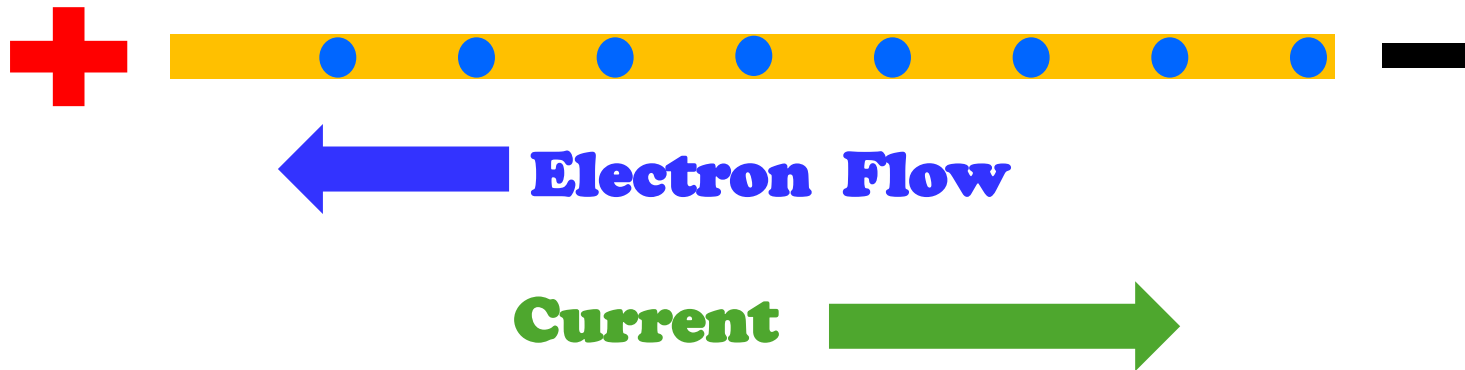


**Direct  
Current**

# Direct Current (DC)



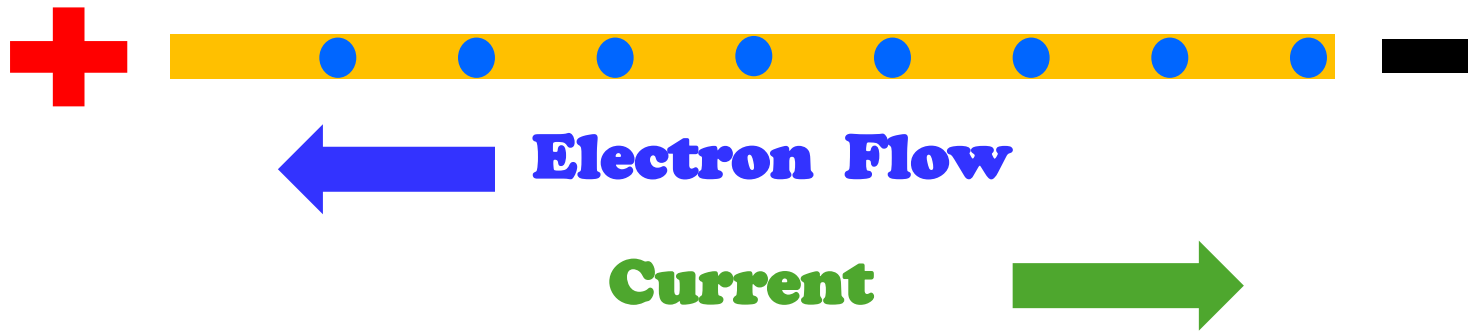
**Current only flow in one direction**



**Ex : Battery**

# Alternating Current (AC)

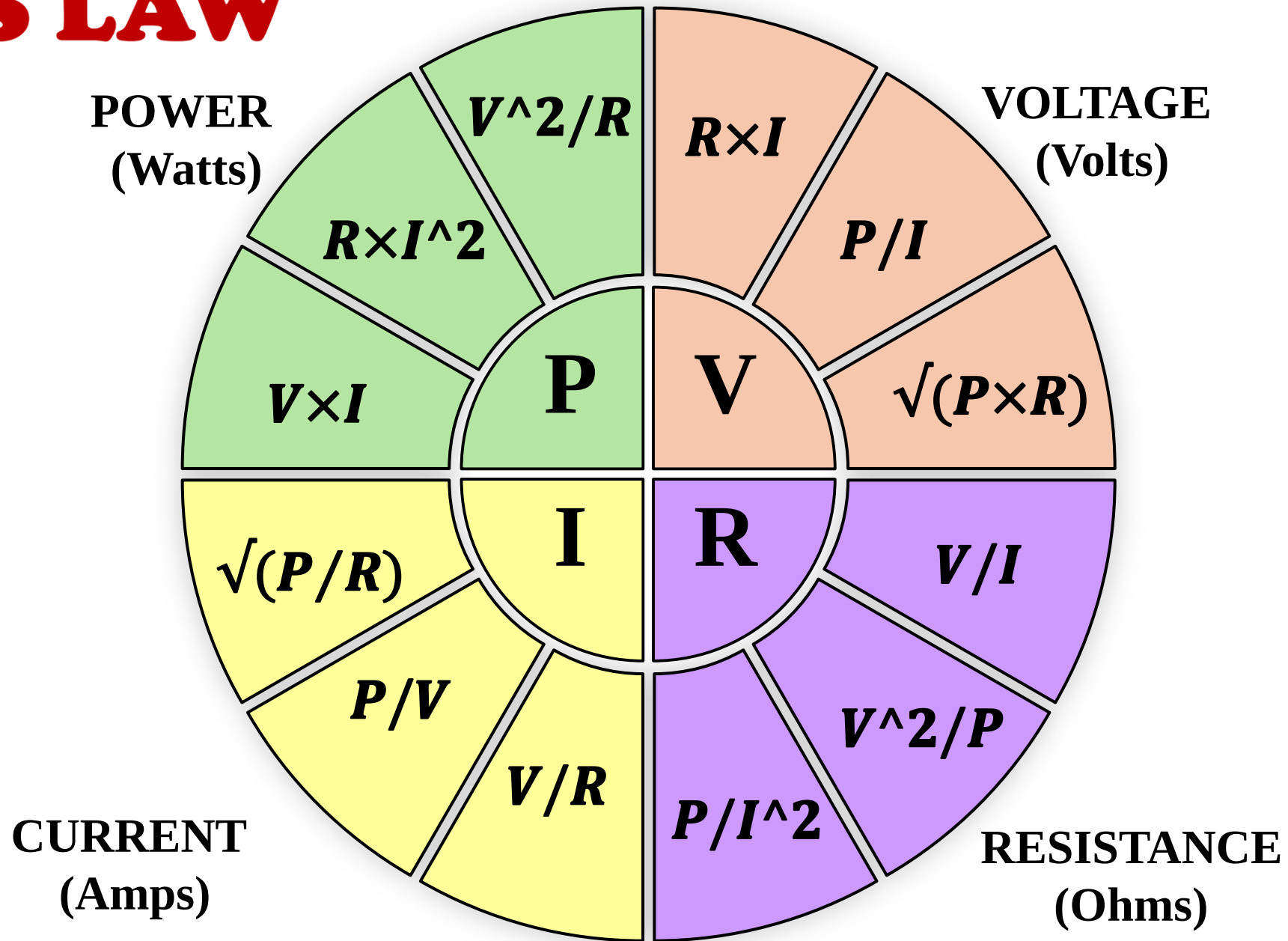
**The direction of Electrons changes rapidly, That's why it is called  
Alternating Current.**



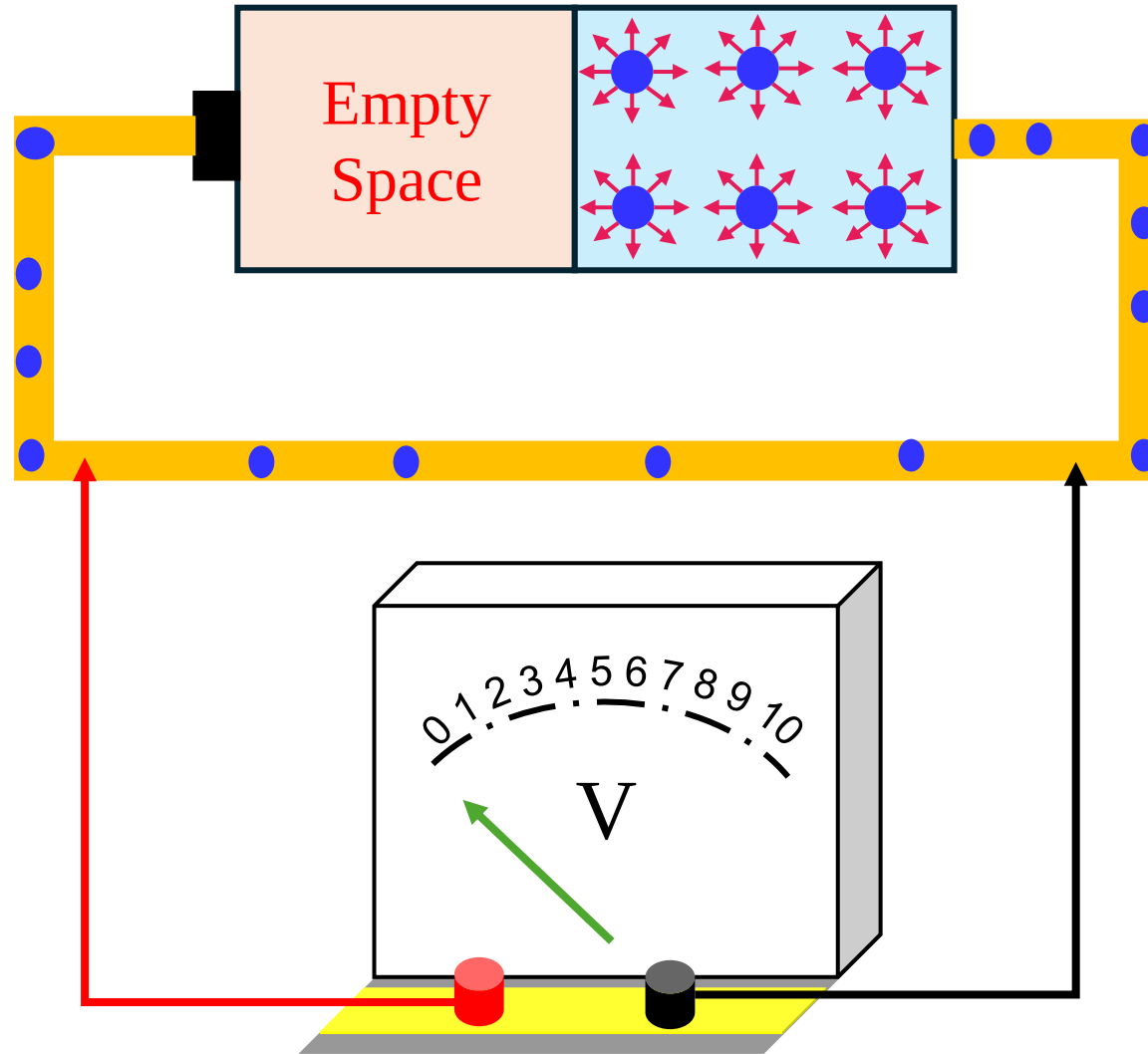
**Ex : Household Electricity**

**Frequency : How many times the current  
switches its direction in 1 sec**

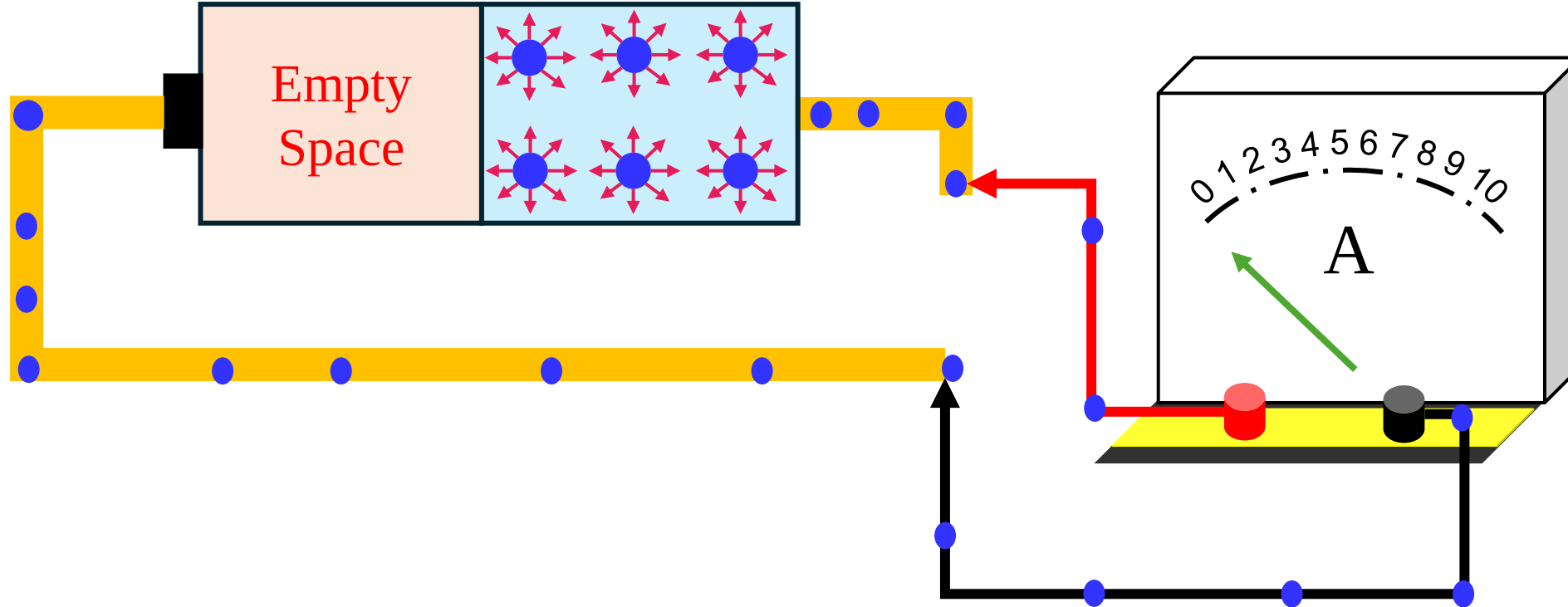
# OHMS LAW



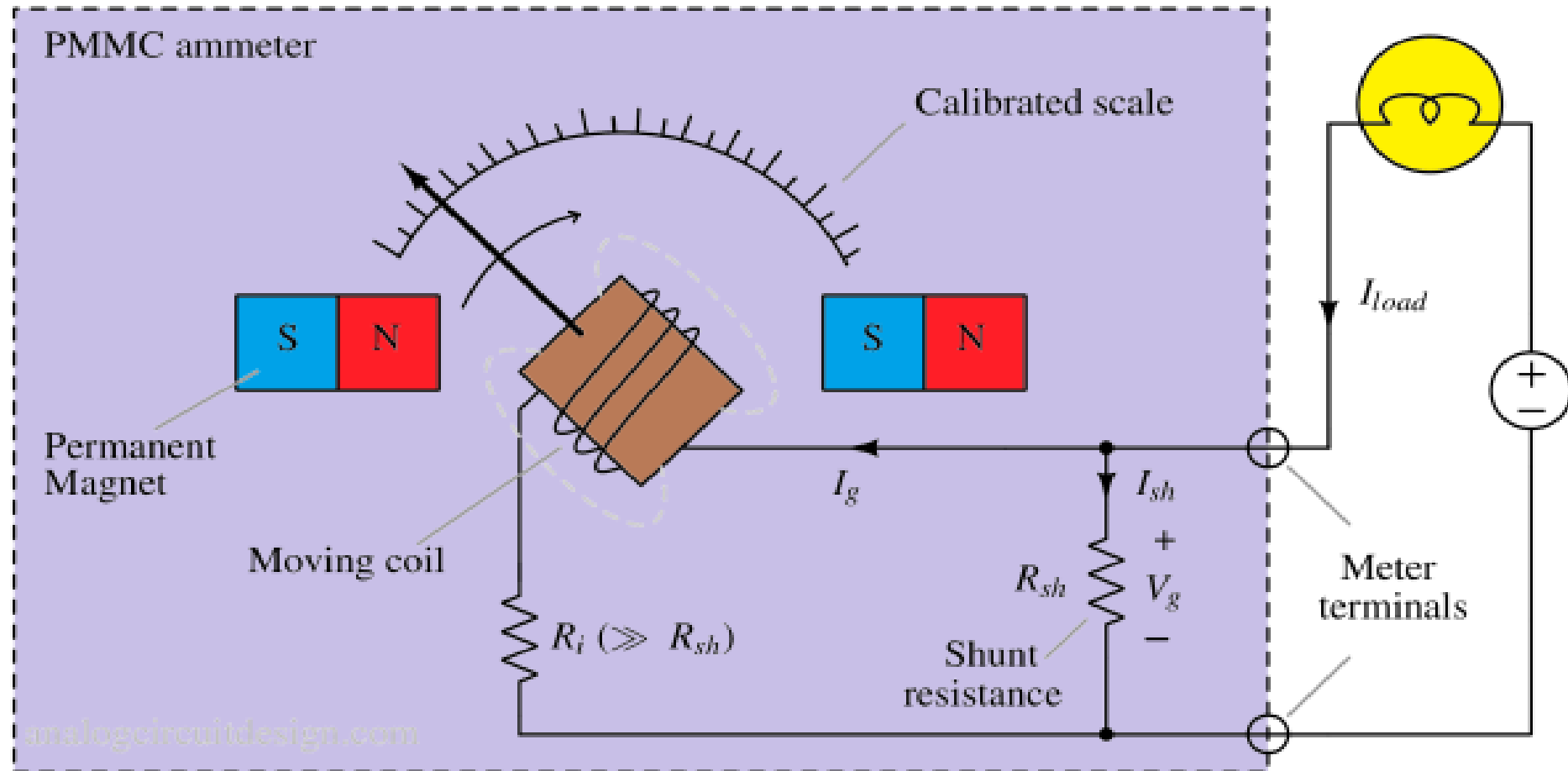
# How to Measure Voltage ?



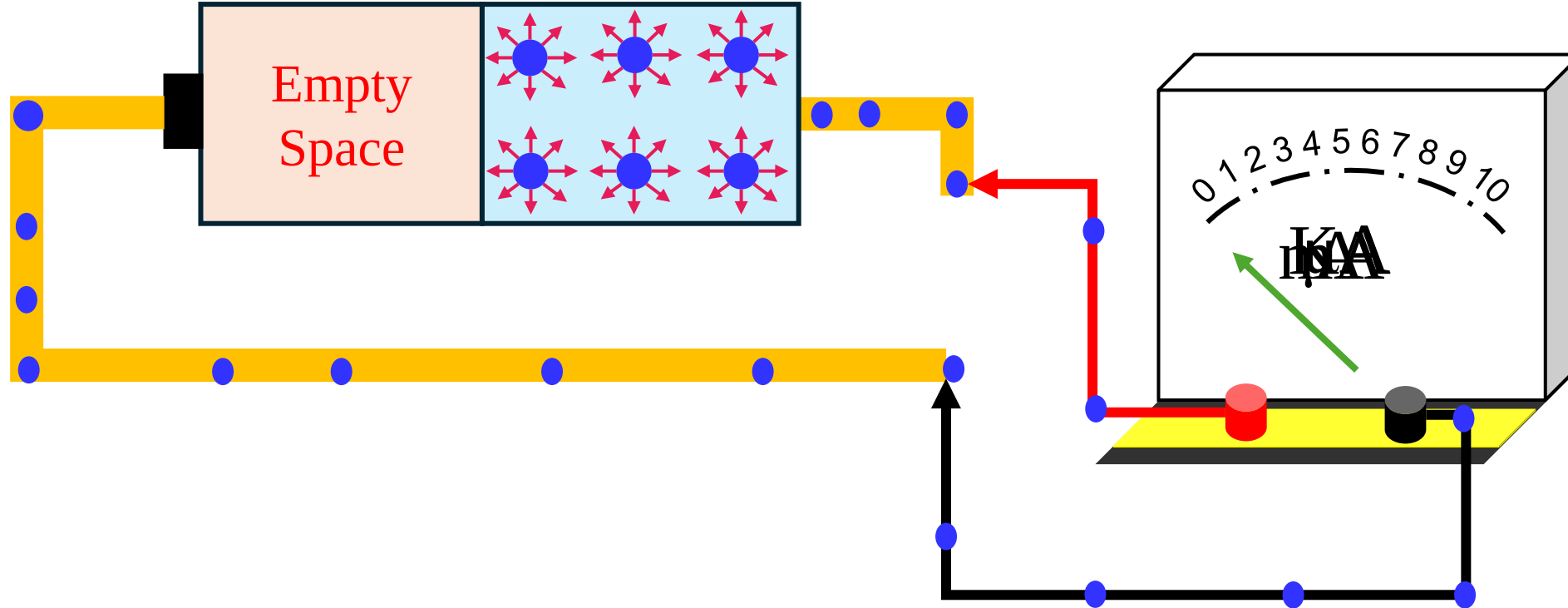
# How to Measure Current ?



# Working Principle of Moving Coil Galvanometer



# Disadvantages of Galvanometer



# Digital Multi-Meter

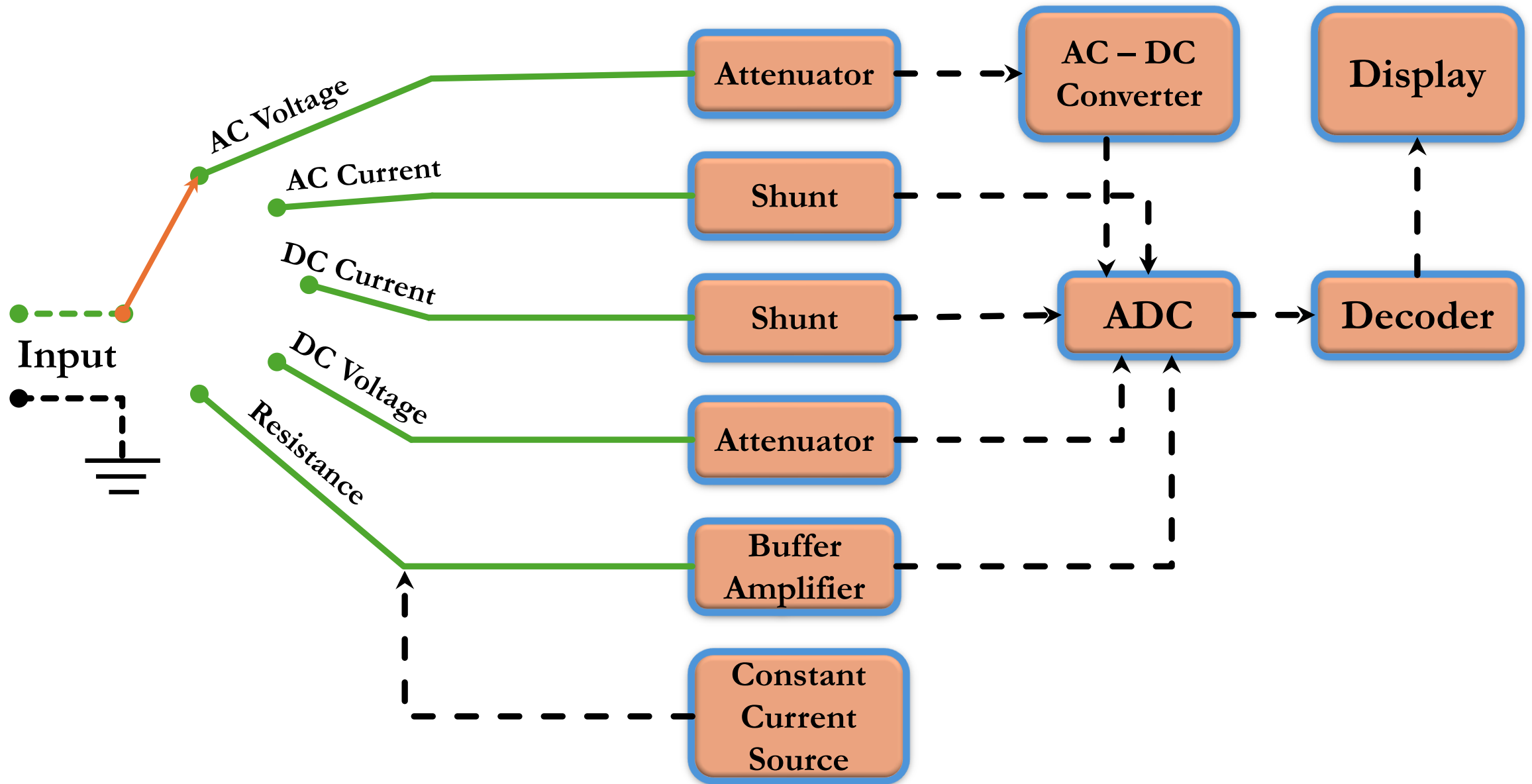


**A Single instrument Helps to measure Multiple parameters and display results in digital form.**

## MEASUREMENT PARAMETERS

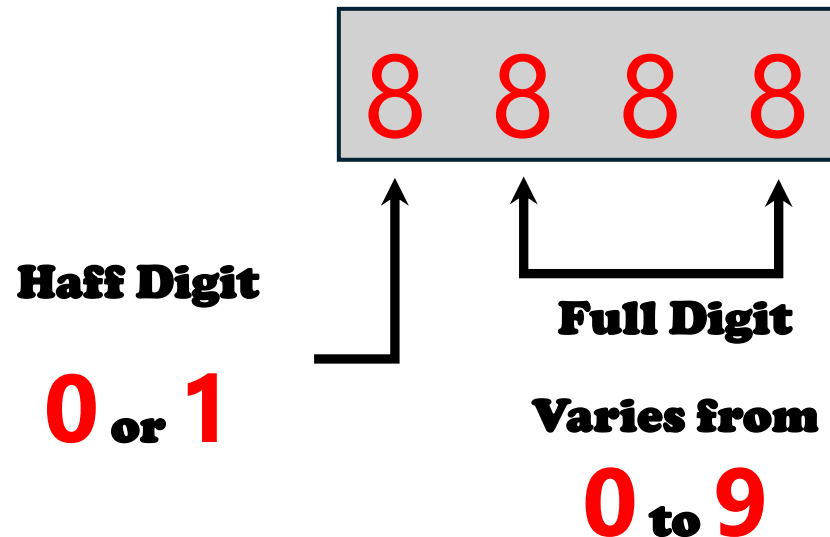
- **AC / DC Voltage**
- **AC / DC Current**
- **Resistance**
- **Capacitance**
- **Frequency**
- **Continuity**
- **Temperature**

# Digital Multi-Meter



# Digital Multi-Meter

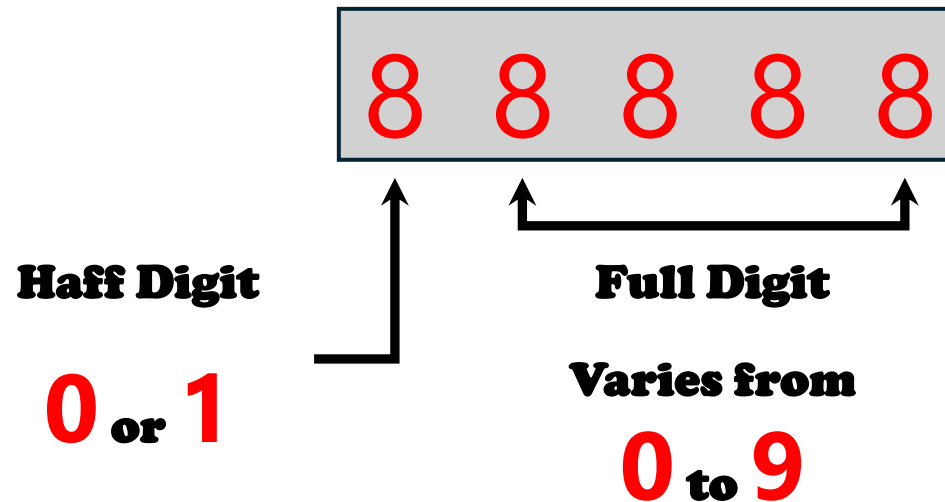
**$3\frac{1}{2}$  Digit “refer to the resolution of the multi-meter's display”**



**Therefore, the range of digital display will be 0 to 1999**

# Digital Multi-Meter

**4½ Digit “refer to the resolution of the multi-meter's display”**



**Therefore, the range of digital display will be 0 to 19999**

# Digital Multi-Meter

**The maximum voltage rating of handheld multi-meters often**  
**600 V or 1000 V**

**Why Handheld Multi-meters are Limited to 600 V / 1000 V**

- **Covers residential, commercial, and industrial low-voltage systems.**
- **Designed to comply with international safety categories (CAT I–IV)**
- **Measuring above 1000 V is dangerous in handheld operation.**

# Bench Digital Multi-Meter

**Benchtop Multi-meters have Higher resolution, Accuracy, stability & precision**



**Half Digit**      **Full Digit**

- Designed for long-duration continuous operation without drift.
- Ideal for R&D, calibration, automated test systems.
- Supports Advanced math functions (statistics, limits, scaling, dB, dBm).
- Support 4-wire resistance (Kelvin) probe measurement.

# **4-wire Kelvin probe measurement**



# Clamp-Meter



**A clamp meter is used to measure current in a conductor without having to make**

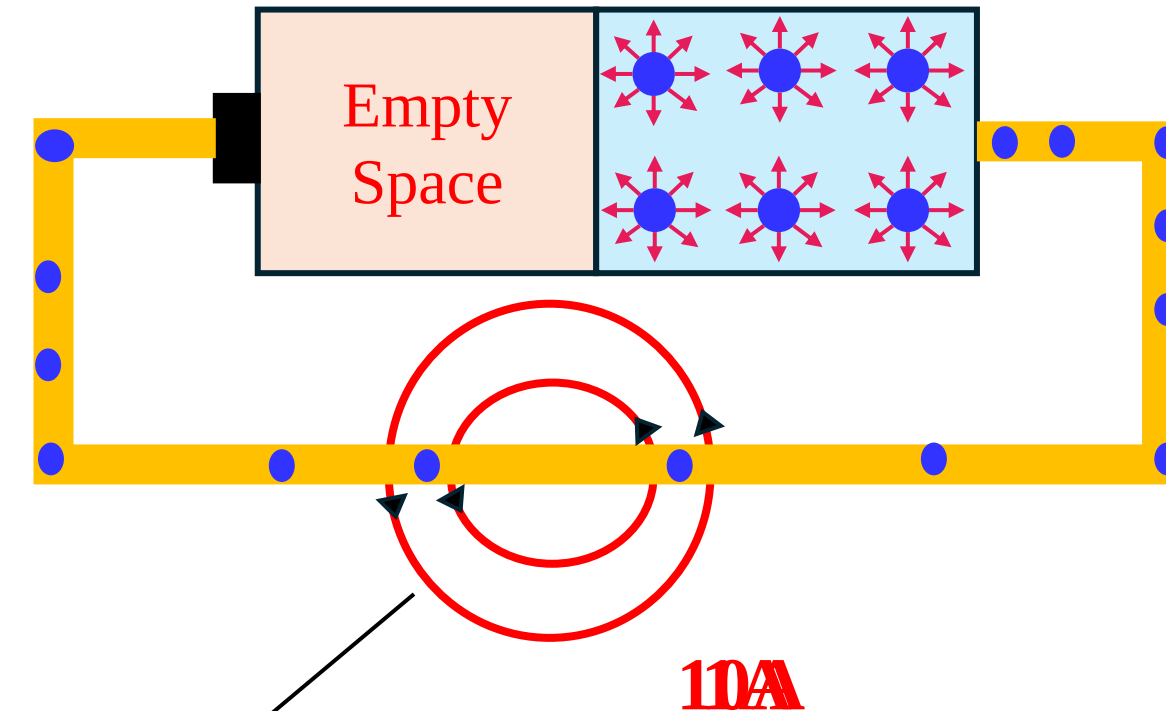
**Direct contact or Disconnect the circuit**

**This non-contact measurement offers advantages in terms of**

- **safety**
- **convenience**
- **efficiency**

**As compared to traditional methods.**

# How to Measure Current ?

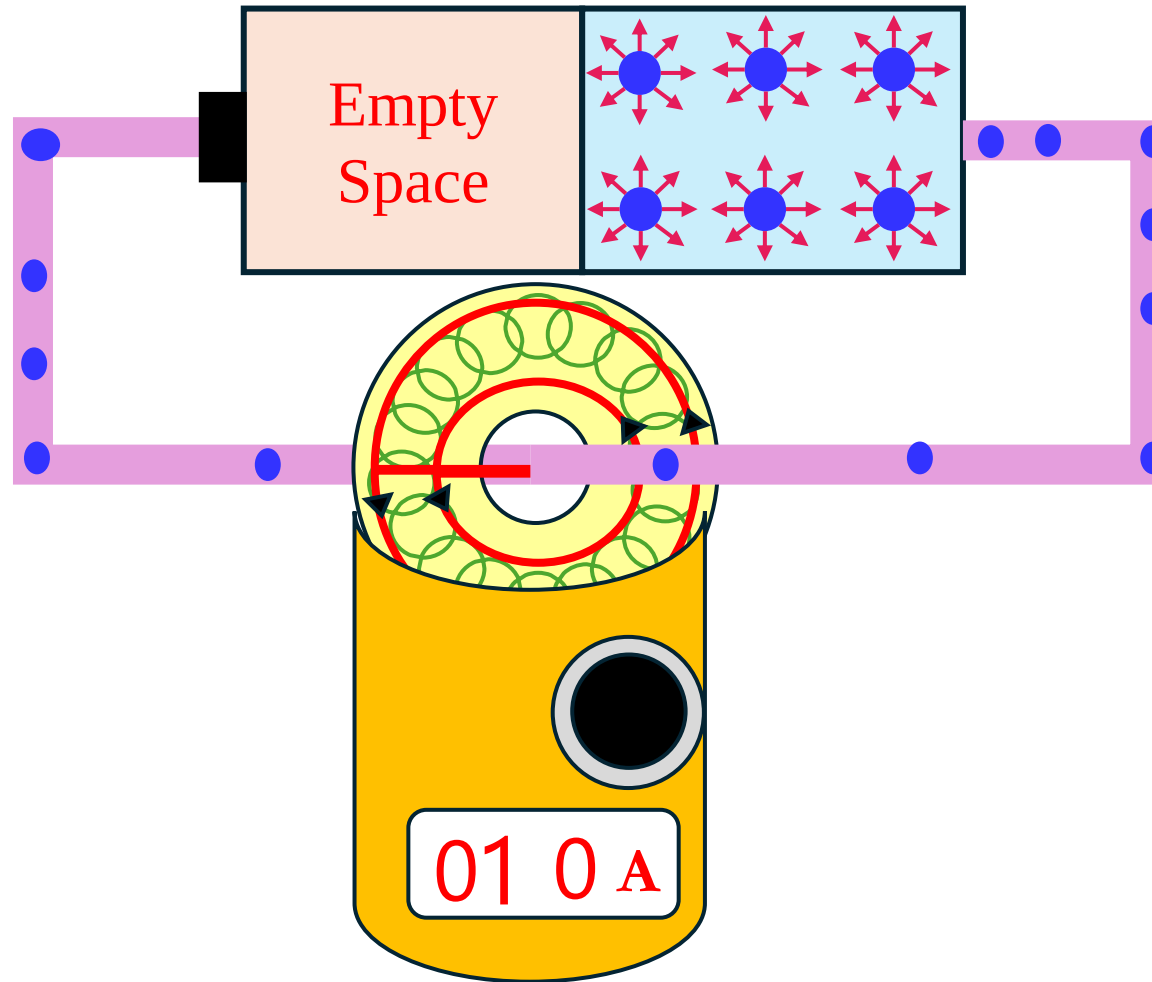


Magnetic  
Fields

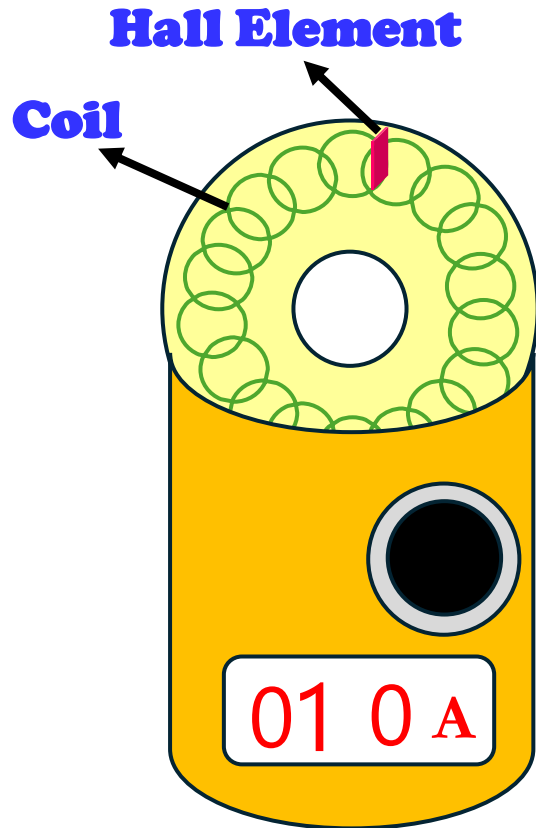
Higher Magnetic Fields = Higher Current

10A

# Current Measurement Using Clamp Meter



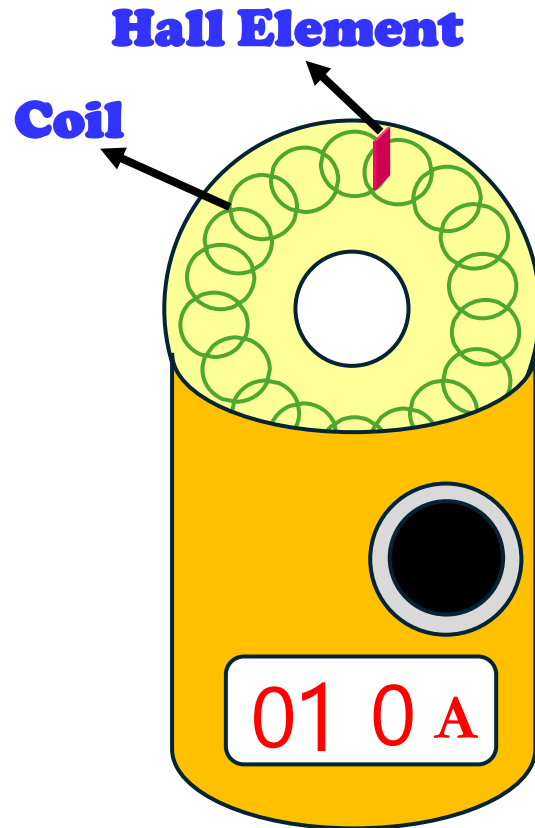
# AC Current Measurement



## Electromagnetic Induction :

- When current flows through a conductor, it produces a magnetic field around it.
- The clamp jaws of the meter act like the core of a current transformer (CT).
- This CT senses the magnetic field and induces a small proportional current in its secondary coil.
- The induced current is then converted into a voltage signal, processed by the meter's electronics, and displayed as the actual current flowing in the conductor.

# DC Current Measurement



## Hall Effect Sensor

- A DC current produces a steady magnetic field (not alternating).
- A Hall-effect sensor is placed in the clamp jaws.
- The Hall sensor detects the magnetic flux density and converts it into a proportional voltage.
- The meter's circuitry processes this voltage and displays the corresponding DC current.

# How to Generate DC Power ?

A **DC power supply** is an electronic device that provides a **stable Direct current (DC) voltage**

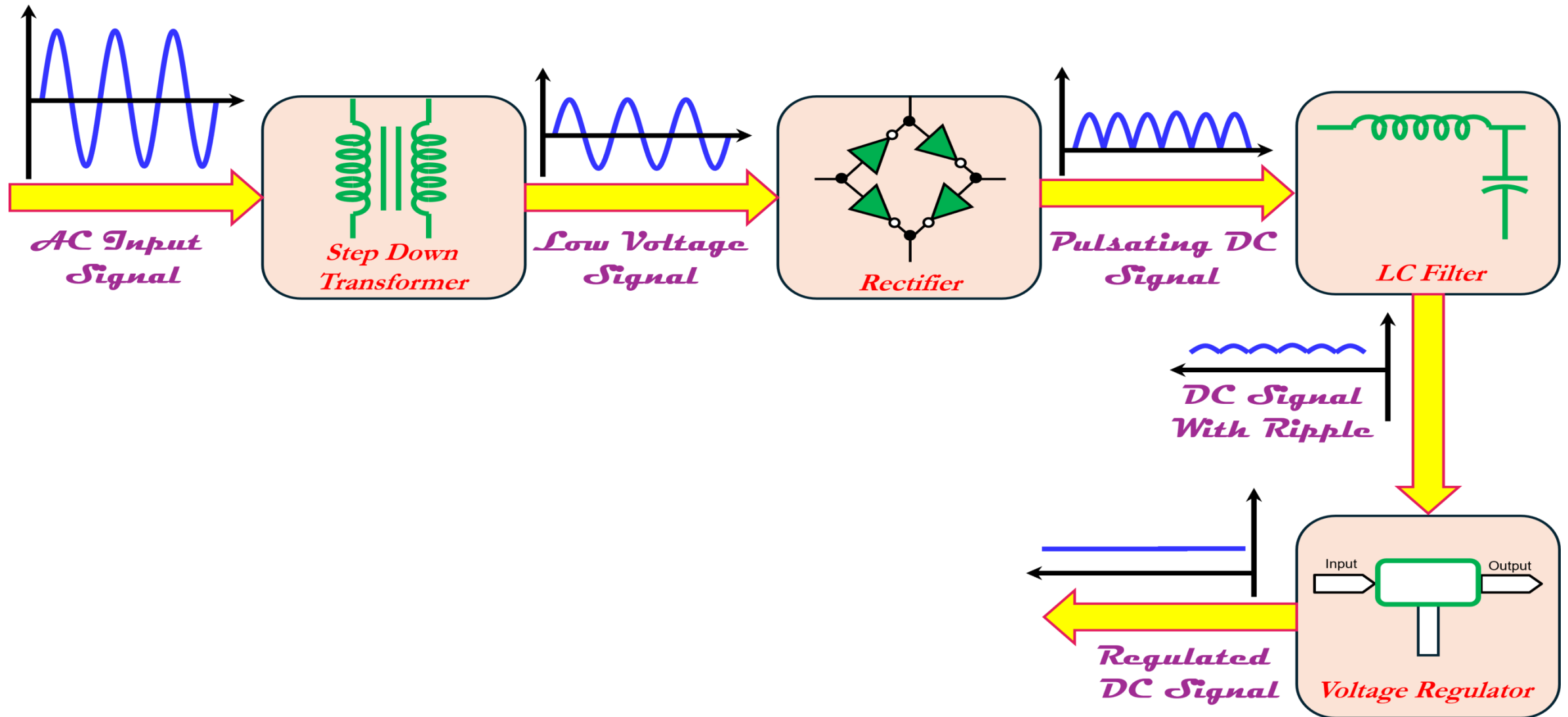
Basically, DC Power Supply is like a very  
“**smart and precise variable battery**”  
that you can fully control, automate, and trust for  
accurate, noise-free DC power output

DC Power Supply **allow us to set a Voltage & Current Limit** within its max Power limit

The Power supply indirectly provides current based on  
what the connected device draws.



# Block Diagram of DC Power Supply



# Modes in DC Power

## Constant Voltage CV-Mode

Maintains a constant output voltage regardless of changes in the load resistance (up to a specified limit).

**Applications:**  
Powering sensitive electronic circuits like microprocessors, communication systems, and IC's that require stable voltage.

## Constant Current CC-Mode

Maintains a constant output current regardless of changes in the load resistance (within certain limits).

**Applications:** Powering devices like LEDs where brightness is current-dependent, charging batteries that require a controlled current flow

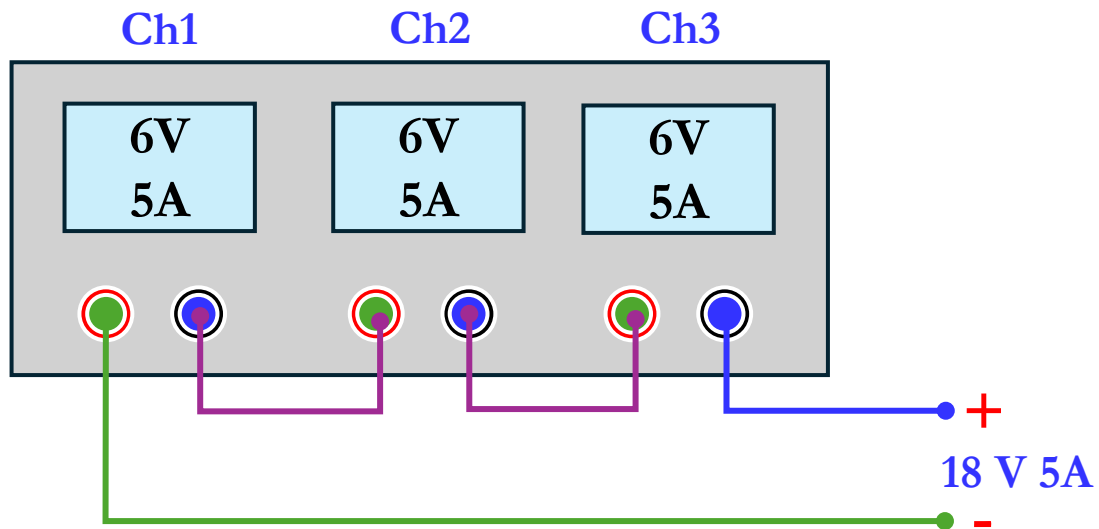
## Constant Power CP-Mode

Maintains a constant output power by adjusting both the voltage and current delivered to the load.

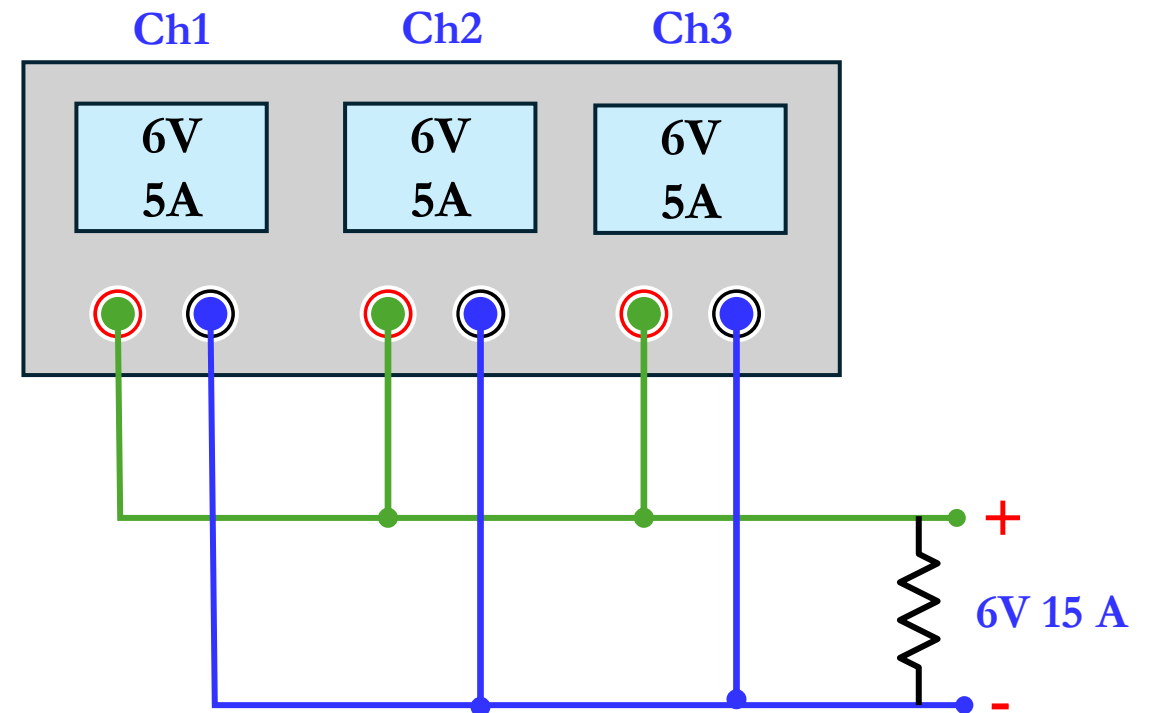
**Applications:** Photovoltaic (PV) testing and simulation, laser applications, and resistive heating elements with varying resistance due to temperature.

# Series & Parallel Connection

## Series Connection For Higher Voltage



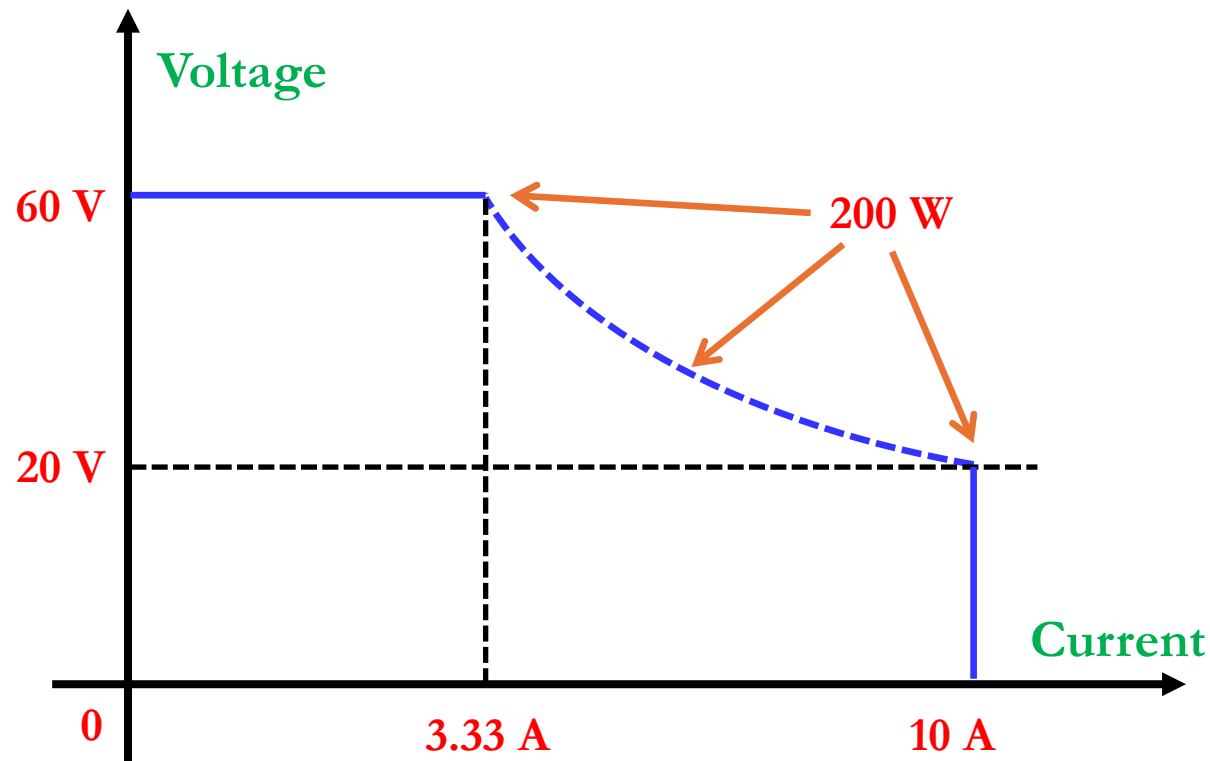
## Parallel Connection For Higher Voltage



# Auto Ranging DC Power Supply

A DC power source that automatically adjusts its voltage and current output to provide the maximum available power within its rated limits for a given load

Keysight E36232A  
60V, 10A – 200W



# How to Generate AC Power ?

The **AC Source** is a device that generates alternating voltage/current, used for powering, simulating, and testing equipment across industries

AC Sources – Allow precise control of:

- Voltage
- Frequency
- Waveform (sine, square, distorted)
- Phase angle



# How to Visualize Signal

An **oscilloscope** is an electronic test instrument that **allows you to see how voltage changes** with time.

An oscilloscope draws a graph of the signal so you can observe its shape, amplitude, timing, & any distortions.

## Key details visible on oscilloscope:

- Amplitude (peak value / RMS equivalent)
- Frequency (Hz)
- Waveform shape (sine, square, noise, distortion, harmonics, etc.)
- Phase difference between voltage and current (useful for AC circuits and power factor analysis).



# Keysight oscilloscope

|                   | 1000 X-Series                                                                     | 2000 X-Series <sup>1</sup>                                                        | HD3 Series                                                                         | 3000G X-Series                                                                      | 4000G X-Series                                                                      | 6000 X-Series                                                                       | EXR-Series                                                                          |
|-------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                   |  |  |  |  |  |  |  |
| Bandwidth         | 50 to 200 MHz                                                                     | 70 to 200 MHz                                                                     | 200 MHz to 1 GHz                                                                   | 100 MHz to 1 GHz                                                                    | 200 MHz to 1.5 GHz                                                                  | 1 GHz to 6 GHz                                                                      | 500 MHz to 6 GHz <sup>3</sup>                                                       |
| Memory (Max)      | 2 Mpts                                                                            | 1 Mpts                                                                            | 100 Mpts/ch                                                                        | 4 Mpts                                                                              | 4 Mpts                                                                              | 4 Mpts                                                                              | 1.6 Gpts                                                                            |
| Sample rate (Max) | 2 GSa/s                                                                           | 2 GSa/s                                                                           | 3.2 GSa/s (per channel)                                                            | 5 GSa/s                                                                             | 5 GSa/s                                                                             | 20 GSa/s                                                                            | 16 GSa/s <sup>3</sup>                                                               |
| Channels          | 2 or 4 analog                                                                     | 2 or 4 analog + 8 digital <sup>2</sup>                                            | 4 analog + 16 digital <sup>2</sup>                                                 | 2 or 4 analog + 16 digital <sup>2</sup>                                             | 2 or 4 analog + 16 digital <sup>2</sup>                                             | 2 or 4 analog + 16 digital <sup>2</sup>                                             | 4 or 8 analog + 16 digital (optional)                                               |
| Display           | 7.0"                                                                              | 8.5"                                                                              | 10.1" capacitive touch gesture-enabled                                             | 8.5" capacitive touch                                                               | 12.1" capacitive touch                                                              | 12.1" capacitive touch                                                              | 15.6" capacitive touch                                                              |
| Update rate       | 200,000 wfms/s on DSO models                                                      | 200,000 wfms/s                                                                    | 1 M wfms/s > 1,300,000 wfms/s                                                      | 1,000,000 wfms/s                                                                    | 1,000,000 wfms/s                                                                    | 450,000 wfms/s                                                                      | > 200,000 wfms/s                                                                    |

*Thank You!*